

외계행성 대기

Exoplanetary Atmospheres

Lecture 2: Observational Techniques (continued) and Definitions Related to Radiation 2026 September 14 (Monday)

Based on: Chapter 1 and 2 of Seager, *Exoplanet Atmospheres* (2010)

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Outline – Lecture 2

Transits and Occultations: Putting It Together

Radial Velocities, Direct Imaging, and Microlensing

Types of Planets

Observational Outlook

Describing Radiation: Intensity and Its Moments

Surface Flux and Observed Flux

Luminosity, Incident Flux, and Blackbodies

Summary

Putting it all together (1): transit and eclipse

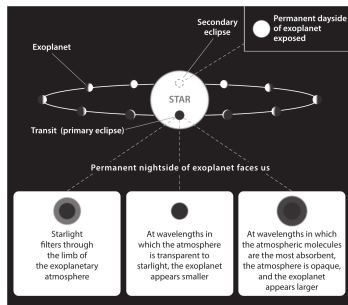


Figure 1: Phases of a transiting exoplanet and the atmosphere's effect on its apparent size during transit. Heng (2017), Fig. 1.4 (upper panel).

This schematic brings together everything discussed so far. It assumes a **tidally locked** planet with permanent dayside and nightside hemispheres — the typical configuration for close-in planets — but the same principles apply more generally.

- ▶ **Transit (primary eclipse):** the nightside faces us and starlight filters through the *limb* of the atmosphere. At wavelengths where the atmosphere is transparent, the planet appears *smaller*; at wavelengths where its molecules absorb most strongly, the atmosphere is opaque and the planet appears *larger*. This is the transmission spectrum.
- ▶ **Secondary eclipse:** the permanent dayside is fully exposed just before and after the planet passes behind the star; subtracting the in-eclipse flux gives the dayside emission spectrum and albedo.
- ▶ In between, the observer sees a changing mix of day and night hemispheres: the phase curve (next slide).

Tidally locked (synchronously rotating): the tidal bulge raised by the star brakes the planet's spin until its rotation period equals the orbital period, so the same hemisphere always faces the star — as the Moon does to Earth. The synchronization time rises steeply with distance (roughly as a^6), so it is only $\sim 10^6$ yr at 0.05 AU but exceeds the age of the system beyond a few tenths of an AU.

Putting it all together (2): phase curve and winds

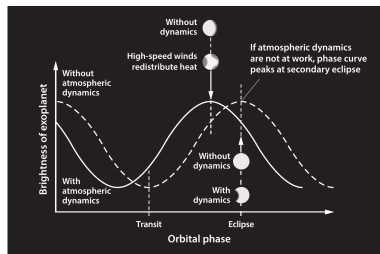


Figure 2: The resulting phase curve, with and without atmospheric dynamics. Heng (2017), Fig. 1.4 (lower panel).

- ▶ **Without atmospheric dynamics:** the hottest point is the *substellar point* (the point on the planet where the star is directly overhead, where the incident flux is greatest), and the brightness distribution is symmetric about it. The observer sees the maximum brightness when the full dayside is in view, so the phase curve *peaks exactly at secondary eclipse*.
- ▶ **With atmospheric dynamics:** high-speed winds (a super-rotating equatorial jet) advect heat away from the substellar point and shift the hot spot in longitude (eastward for hot Jupiters). Longitudes east of the substellar point face the observer *before* eclipse, so the brightness peak occurs *before* secondary eclipse.
- ▶ A measured **peak offset therefore implies winds** (Showman & Guillot 2002). For HD 189733b the $8\text{ }\mu\text{m}$ peak leads the eclipse by $\sim 16^\circ$ in longitude, and the day–night contrast is modest — efficient heat redistribution (Knutson et al. 2007).

Radial velocity measurements

The Sun and its planets orbit a common center of mass. As the star wobbles, its light is blue- and redshifted; the periodic Doppler shift produces the **radial-velocity** (RV) signal — the method by which 51 Peg b was discovered. The semi-amplitude of the stellar velocity is

$$K_* = \left(\frac{2\pi G}{t_{\text{period}}} \right)^{1/3} \frac{M_p \sin i}{(M_* + M_p)^{2/3}} (1 - e^2)^{-1/2}$$

where i is the orbital inclination and e is the eccentricity.

- ▶ **The $M_p \sin i$ degeneracy:** RV alone gives a *minimum mass*. M_* comes from stellar-evolution models. If the planet *transits*, $\sin i \simeq 1$, allowing its true mass to be determined. Transit (R_p) + RV (M_p) $\Rightarrow$ bulk density and surface gravity $g = GM_p/R_p^2$ — essential inputs for interpreting any spectrum ($H = k_B T/mg$).
- ▶ **Magnitudes:** a hot Jupiter produces $K_* \sim 0.1 \text{ km s}^{-1}$ (an order of magnitude faster than a sprinter); Jupiter (12-yr orbit) $\sim 10 \text{ m s}^{-1}$; Earth $\sim 10 \text{ cm s}^{-1}$ (humans walk at $\sim 1 \text{ m s}^{-1}$). The technological demand increases steeply toward lower planetary masses.
- ▶ **Rossiter–McLaughlin effect:** a rotating star has a blueshifted approaching hemisphere and a redshifted receding hemisphere. A transiting planet blocks one hemisphere and then the other, producing an anomalous RV signal whose shape gives the sky-projected *spin–orbit misalignment*. Many hot Jupiters are misaligned — a clue to their migration history.

High-resolution spectroscopy of atmospheres

Ground-based spectrographs with resolution $\lambda/\Delta\lambda \sim 10^5$ (versus 10^2 – 10^3 for space-based low-resolution spectra) can also study atmospheres:

- ▶ The planet's own lines (in transmission or emission) shift by tens of km s^{-1} during the observation because of the planet's orbital motion, whereas stellar and **telluric** lines (from Earth's atmosphere) do not. This separates the three components.
- ▶ The thousands of individual molecular lines are too weak to see one by one; **cross-correlating** the residual spectrum with model templates of CO, H₂O, CH₄, ... stacks them into a detectable signal (Snellen et al. 2010; Brogi et al. 2012).
- ▶ This method yields unambiguous molecular identifications, the planet's orbital velocity (hence its true mass without $\sin i$), line Doppler shifts from winds, and even rotation.
- ▶ High-resolution *transmission* spectra can also resolve individual strong lines (Na, He 1083 nm, H α) that probe the upper atmosphere and atmospheric escape.

Low-resolution space spectroscopy and high-resolution ground-based spectroscopy are complementary: the former measures absolute band strengths, whereas the latter measures line shapes and velocities.

Direct imaging

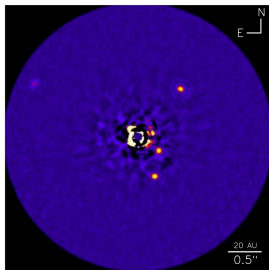


Figure 3: The four young gas-giant planets of HR 8799 (Marois et al. 2008, 2010), shown after starlight suppression. Heng (2017), Fig. 1.5.

The goal is to image the planet directly and obtain its spectrum.

- ▶ The planet-to-star flux ratio is typically 10^{-6} – 10^{-9} (infrared to visible) for mature planets. For **young** (10^7 – 10^8 yr) planets still glowing with residual heat from their formation, the contrast can be as favorable as 10^{-4} .
- ▶ The starlight must be suppressed (coronagraph, adaptive optics, angular differential imaging) to isolate the planet and measure its emission spectrum $F(\lambda)$.
- ▶ HR 8799 b, c, d, e: 5 – $10 M_J$ planets at 15 – 70 AU; spectroscopy revealed H_2O and CO in their atmospheres (Barman et al. 2011; Konopacky et al. 2013).

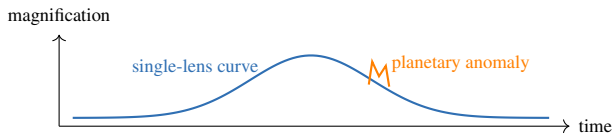
Drawback: neither the radius nor the mass is measured directly; both are inferred from spectral and evolutionary models, which depend on the uncertain age and initial entropy, so they are *model-dependent*.

Strength: direct imaging probes planets at wide separations, complementing the close-in transiting population. With future space coronagraphs and starshades, it also offers a path toward imaging Earth twins.

Gravitational microlensing

Paczynski (1986) proposed that unseen objects in the Galactic halo act as gravitational lenses, causing the transient brightening of a background star. He estimated the probability of such an event to be one in a million, implying that roughly a million stars must be monitored to detect one event. Mao & Paczynski (1991) applied the same reasoning to planet detection.

- ▶ A foreground (lens) star passing in front of a background (source) star produces a smooth, symmetric rise and fall of the magnified light as the two move past each other. (Stars in our Galaxy are not static: they move at $\sim 10\text{--}100\text{ km s}^{-1}$.)
- ▶ If a planet orbits the lens star, it distorts this smooth curve — a short-lived anomaly lasting hours to days, “in a manner analogous to astigmatism.”
- ▶ The shape of the anomaly yields the planet-to-star mass ratio and projected separation (in units of the Einstein radius).



Microlensing: the Einstein radius and limitations

The characteristic length scale at which lensing occurs is the **Einstein radius**, the geometric mean of the Schwarzschild radius of the lens and the distance to the lens:

$$l_{\text{Einstein}} \sim \frac{(GM_* l_{\text{lens}})^{1/2}}{c}, \quad l_{\text{Schwarzschild}} \sim \frac{GM_*}{c^2}$$

where M_* is the mass of the lens star and l_{lens} its distance from the observer. (This is the order-of-magnitude form; the exact expression also involves the distance to the source and the lens–source separation.)

- ▶ For a Sun-like lens at kpc distances, l_{Einstein} is a few AU. The numerical coincidence between this scale and the orbital distances of Jupiter-like planets ($a \simeq 5$ AU) makes the probability of detecting the planetary anomaly relatively high ($\sim 10\%$) once a stellar event is under way (Gould & Loeb 1992).
- ▶ **Limitations.** Microlensing detects planets (mass ratio, projected separation) but cannot characterize their atmospheres; the detection is *unrepeatable*; and the lens star is typically hard to distinguish photometrically from the source star, so determining the planet's absolute mass requires additional information.
- ▶ **Strength.** Sensitivity to cold planets beyond the snow line, to low-mass planets, and to free-floating planets, at kpc distances toward the Galactic bulge — a demographic census that complements transits and radial velocities (the Roman Space Telescope microlensing survey).

Techniques compared

Technique	Measured quantities	Atmospheric potential	Main limitation / bias
Transit & eclipse	R_p/R_* , i , ρ_* , a/R_* ; F/F_* , A_g , T_{day}	High: transmission, emission, phase curves, eclipse maps	Needs edge-on geometry ($P \simeq R_*/a$); favors close-in planets
Radial velocity	$M_p \sin i$, e , t_{period} ; spin-orbit angle	Indirect: masses and g ; high-resolution cross-correlation of molecular lines	Favors massive, close-in planets around quiet, slowly rotating stars
Direct imaging	Flux and emission spectrum $F(\lambda)$, astrometric orbit	Medium-high for young, hot giants; the route to Earth twins	Extreme contrast; wide separations only; M_p , R_p model-dependent
Microlensing	M_p/M_* , projected separation	None	Transient, unrepeatable; faint, distant hosts

Transit + RV is the most powerful combination: radius *and* mass give density, composition class, and surface gravity, which every atmospheric model needs.

The diversity of exoplanets

One of the most striking features of the exoplanet population is its **remarkable diversity**: planets span a nearly *continuous* range of masses and semimajor axes.

- ▶ Planet formation is *stochastic*: planets of many masses form at many locations in the protoplanetary disk.
- ▶ *Migration* carries planets very close to the star.
- ▶ The sloping lower envelope of the RV points is a *selection effect* ($K_* \propto M_p a^{-1/2}$); the mass of pulsar planets is known from timing.

We should therefore expect the diversity of *atmospheres* to rival diversity of masses and orbits. Today (2026) the sample exceeds 6000 confirmed planets, dominated by *Kepler/TESS* transits, and the “desert” below the RV envelope has filled in.

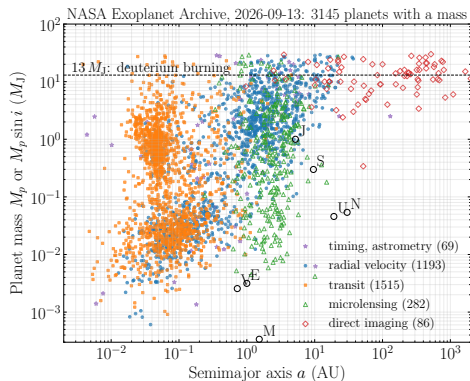


Figure 4: Known exoplanets with a measured mass (or $M_p \sin i$), by discovery technique. Solar System planets are marked by their initials. Compare Seager (2010), Fig. 1.5, which showed the sample as of January 2010.

The Solar System menagerie

Despite their different appearances, Solar System planets fall into two main categories.

Terrestrial planets — Mercury, Venus, Earth, Mars.

- ▶ Composed predominantly of rock and metals, with thin atmospheres.
- ▶ Atmospheres are *evolved*: escape of light gases and gas–surface reactions (weathering, ocean dissolution) have changed them substantially from their primitive state.
- ▶ Venus: 90 bar CO₂; Earth: 1 bar N₂/O₂; Mars: 6 mbar CO₂ — three very different outcomes.

Gravity provides the dividing line: massive planets retain nearly everything, including H₂ and He; light planets lose their light gases and keep only what is outgassed or supplied later.

Giant planets — Jupiter, Saturn, Uranus, Neptune.

- ▶ They have no rocky surface and can reach hundreds of Earth masses.
- ▶ *Gas giants* (Jupiter, Saturn): almost entirely H and He, massive atmospheres, liquid metallic-hydrogen interiors.
- ▶ *Ice giants* (Uranus, Neptune): mostly water, methane, and ammonia “ices,” with a 1–2 $M_{\oplus}$ H/He envelope.
- ▶ Their atmospheres are *primitive*: they have undergone relatively little evolution and therefore retain approximately the gases with which they formed (solar-like H/He with enhanced heavy elements).

Definitions (after Levine, Shaklan & Kasting 2006)

- ▶ **Planet:** a gravitationally bound object that is supported against collapse by *electron degeneracy pressure* or *Coulomb pressure*, orbits a star, and *never* sustains nuclear fusion in its core during its lifetime. Theory places this limit at $M_p \lesssim 13 M_J$ (deuterium burning). No convincing lower mass limit exists.
- ▶ **Exoplanet:** a planet orbiting a star other than the Sun (sometimes also used for free-floating planets without a host).
- ▶ **Terrestrial planet:** supported primarily by Coulomb pressure, with a *surface* defined by the radial extent of the solid interior or liquid outer layer (“rocky planet”); a thin gas atmosphere may exist above it.
- ▶ **Giant planet:** an object whose mass substantially exceeds that of terrestrial planets but remains below that of brown dwarfs, $0.03 M_J < M_p < 13 M_J$ ($\sim 10 M_\oplus < M_p \lesssim 4000 M_\oplus$); interiors contain a substantial fraction of metallic hydrogen around a rocky core.
- ▶ **Brown dwarf:** $13\text{--}80 M_J$; burns deuterium (and, above $\sim 65 M_J$, lithium) but never hydrogen. Their atmospheres are physically continuous with those of giant planets.

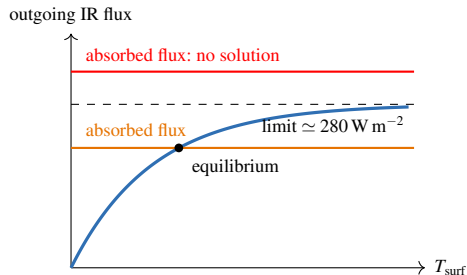
Habitability-related definitions

- ▶ **Habitable zone (HZ):** the region around a star in which a planet may maintain *liquid water on its surface*.
 - ▶ *Empirical* boundaries: Venus appears to have lost its water long ago (inner edge $\lesssim 0.75$ AU for the young Sun); Mars appears to have had surface water early on (outer edge $\gtrsim 1.5$ AU).
 - ▶ *Model* boundaries: runaway greenhouse (inner) and maximum CO₂ greenhouse (outer), e.g. Kasting et al. (1993), Kopparapu et al. (2013) — the mechanisms are on the next slide.
- ▶ **Potentially habitable planet:** one whose orbit lies within the HZ.
- ▶ **Habitable planet:** a terrestrial planet on whose surface liquid water exists in steady state. This presumes that life, like life on Earth, requires liquid water — and that the water and any life must be *at the surface* to be detectable remotely.
- ▶ **Earth-like planet:** similar to Earth in mass, radius, and temperature.
- ▶ **Earth twin:** an Earth-like planet with liquid-water oceans and continental land masses.

Being in the HZ is necessary but far from sufficient: Venus is near the HZ and the Moon is inside it. The atmosphere decides.

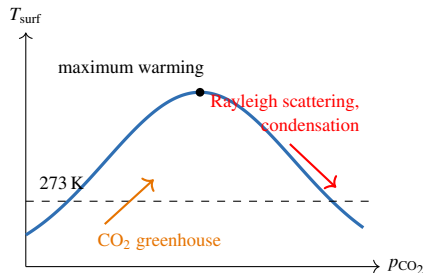
What sets the two edges of the habitable zone

Inner edge: the runaway greenhouse



More starlight evaporates more water, itself a strong greenhouse gas, so a wetter atmosphere radiates from higher, colder layers and the outgoing infrared flux *saturates*. Past the limit no surface temperature can balance the absorbed starlight: the ocean enters the atmosphere and its hydrogen escapes — Venus's likely history.

Outer edge: the maximum CO₂ greenhouse



A cooling planet accumulates CO₂, which warms it back. Beyond a critical column, however, Rayleigh scattering of the incoming starlight and CO₂ condensation take over and cause net cooling. The warming therefore reaches a maximum: where that maximum no longer reaches 273 K, no amount of CO₂ keeps the surface above freezing.

Schematic. The saturation of the outgoing infrared flux is the Simpson–Nakajima limit (Nakajima, Hayashi & Abe 1992; Goldblatt & Watson 2012); the CO₂ maximum follows Kasting, Whitmire & Reynolds (1993) and Kopparapu et al. (2013).

Planets without Solar System counterparts

Class	Mass range	Composition	Atmosphere
Super-Earth	$1\text{--}10 M_{\oplus}$	predominantly rocky (loosely defined)	thin, secondary (outgassed), or none
Mini-Neptune	$\sim 10\text{--}15 M_{\oplus}$	rock/ice core with gas envelope	H/He envelope, volatile-rich
Water world	—	$\geq 25\%$ water by mass	steam or H_2O -dominated; deep oceans
Carbon planet	—	more carbon than silicon (C/O $\gtrsim 1$ in the disk)	CO -, CH_4 -, hydrocarbon-rich

These terms are loosely defined and overlap in part. Super-Earths are intended to include predominantly rocky planets; mini-Neptunes are planets of roughly $10\text{--}15 M_{\oplus}$ with gas envelopes; water worlds and carbon planets are defined by bulk composition rather than by mass.

Super-Earths, mini-Neptunes, and the degeneracy problem

- ▶ Super-Earths and mini-Neptunes are the *most common* planets found by *Kepler*, yet they are absent from the Solar System. Their existence was one of the major surprises of the transit surveys.
- ▶ The boundary between them (rocky versus gas-enveloped) is now known to form a “radius valley” near $1.5\text{--}2 R_{\oplus}$: planets below it have lost their primordial H/He envelopes, whereas those above it have retained a few percent by mass of H/He, which inflates the radius considerably (Fulton et al. 2017).
- ▶ Observations *cannot always distinguish between* these planet types. A given mass and radius can be matched by different mixtures of iron, rock, water, and H/He — the well-known degeneracy of interior models. A $2 R_{\oplus}$, $6 M_{\oplus}$ planet may be a rocky core with a thin H/He envelope, or a water-rich world with a steam atmosphere.
- ▶ The *atmosphere* is often the crucial diagnostic: a hydrogen-dominated atmosphere has a large scale height and strong transmission features, whereas a heavy (H_2O , CO_2 , N_2) secondary atmosphere has a small scale height and a nearly featureless transmission spectrum. This is why the mean molecular mass m in $H = k_{\text{B}}T/mg$ matters so much.

Where do atmospheres come from?

There is no definitive classification of *atmosphere* types yet, but we expect a diversity as wide as that of orbits, masses, and radii.

Two origins.

- ▶ **Direct capture** of gas from the protoplanetary disk — for massive planets (Jupiter, Saturn): a *primary* atmosphere of H_2 and He with disk composition.
- ▶ **Outgassing** during and after accretion — for low-mass planets (Earth, Venus): a *secondary* atmosphere of H_2O , CO_2 , N_2 , ... released from the solid material.

Sources versus sinks. The final atmospheric mass and composition reflect the balance between

- ▶ *sources*: captured or outgassed gas — depending on the planet's *location* in the disk during formation, because of the disk's compositional gradients (snow lines of H_2O , CO_2 , CO);
- ▶ *sinks*: atmospheric *escape* (thermal, hydrodynamic, non-thermal), *sequestration* (gas leaving the atmosphere by being locked away in oceans and rock, e.g. CO_2 into carbonates), and photochemical destruction.

The full range of atmospheric compositions remains to be uncovered, both theoretically and observationally. The formalism developed in this course is meant to be general enough to interpret *any* of them.

Future missions and telescopes

Although exact mission configurations remain uncertain, the broad priorities are clear:

- ▶ **Space-based spectroscopy** from the UV to the infrared to build statistical samples of atmospheres and search for trends: HST (UV–near-IR) and **JWST** (0.6–28 μm , since 2022); **Ariel** (ESA, launch $\sim$ 2029) will survey $\sim$ 1000 transiting planets in transmission and emission.
- ▶ **Bright, nearby targets:** *CHEOPS* (2019), *TESS* (2018), and *PLATO* ($\sim$ 2026) greatly enlarge the sample of transiting planets around bright stars, the prerequisite for atmospheric follow-up.
- ▶ **Ground:** ever-larger mirrors (the 30–40 m-class ELTs) equipped with high-resolution spectrographs and extreme adaptive optics to reach fainter sources and detect reflected light from nearby temperate planets.
- ▶ **Ultimately:** a space coronagraph or starshade (Habitable Worlds Observatory concept), or a fleet of formation-flying telescopes doing *interferometry*, to image an Earth twin and take its spectrum — the pale blue dot, again.

Chapter 1 wrap-up: techniques and planet types

Other tools. Radial velocities give $M_p \sin i$ — the true mass when the planet transits — together with the eccentricity and, through the Rossiter–McLaughlin effect, the spin–orbit angle; at high resolution they also detect individual molecules by cross-correlation. Direct imaging gives spectra of young, self-luminous giants at wide separations. Microlensing detects cold and low-mass planets but cannot characterize their atmospheres.

The zoo. Planets span a continuum of masses and orbits: terrestrial, giant, super Earth, mini-Neptune, water, and carbon worlds. Atmospheres are either captured from the disk or outgassed from the interior, and are then shaped by escape and by sequestration in oceans and crust. Expect a diversity of atmospheres at least as wide as the diversity of masses and orbits.

What comes next. Everything discussed so far has concerned *observations*. To interpret these data, we need a precise language for radiation — intensity, flux, and their moments — the subject of the rest of this lecture.

References (Chapter 1)

- ▶ Seager, S. 2010, *Exoplanet Atmospheres: Physical Processes* (Princeton University Press), Chapter 1.
- ▶ Heng, K. 2017, *Exoplanetary Atmospheres: Theoretical Concepts and Foundations* (Princeton University Press), Chapter 1.
- ▶ Showman, A. P. and Guillot, T. 2002, *A&A* **385**, 166; Knutson, H. A., et al. 2007, *Nature* **447**, 183 (phase curves and winds).
- ▶ Marois, C., et al. 2008, *Science* **322**, 1348; Kalas, P., et al. 2008, *Science* **322**, 1345 (direct imaging).
- ▶ Snellen, I. A. G., et al. 2010, *Nature* **465**, 1049; Brogi, M., et al. 2012, *Nature* **486**, 502 (high-resolution cross-correlation).
- ▶ Mao, S. and Paczyński, B. 1991, *ApJ* **374**, L37; Gould, A. and Loeb, A. 1992, *ApJ* **396**, 104 (microlensing).
- ▶ Levine, M., Shaklan, S., and Kasting, J. 2006, *Terrestrial Planet Finder STDT Report*, NASA/JPL D-34923 (definitions).
- ▶ Fulton, B. J., et al. 2017, *AJ* **154**, 109 (the radius valley).

Why the definitions matter

The goal of the course is to understand the origin of a planetary spectrum and the physical processes that shape it: the gases and solid particles present, the planetary albedo, and the vertical temperature structure. Everything starts from a precise description of the *radiation* traveling through the atmosphere.

- ▶ The terms **intensity**, **surface flux**, and **flux at Earth** are used differently in other books and in the literature. We adopt one consistent set of definitions and use it throughout (Seager's conventions).
- ▶ The same words hide subtleties: is the flux a vector or a scalar? Over which solid angle is it integrated? At the planet or at the detector? Is the “surface” a solid surface?
- ▶ In this chapter, “**surface**” means either the solid surface of a planet or the atmospheric layers from which the radiation emerges (the photosphere, in stellar astrophysics).

With these definitions in hand, we can define planetary temperatures, albedos, and planet-star flux ratios (Chapter 3), and later write down the equation of radiative transfer (Chapter 5).

The specific intensity I

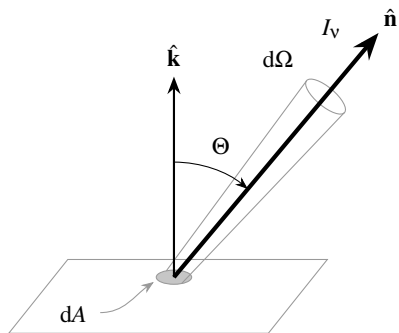


Figure 5: Definition of the specific intensity $I(\mathbf{x}, \hat{\mathbf{n}}, \nu, t)$.
After Seager (2010), Fig. 2.1.

Radiation is energy carried by photons. We describe the energy carried by photons in a single *beam* using the intensity I . It is the energy passing through an area dA , within a solid angle $d\Omega$ centered on the direction $\hat{\mathbf{n}}$, per frequency interval, per unit time:

$$dE(\nu, t) = I(\mathbf{x}, \hat{\mathbf{n}}, \nu, t) \hat{\mathbf{n}} \cdot \hat{\mathbf{k}} d\Omega dA d\nu dt$$

- ▶ SI units: $\text{J m}^{-2} \text{sr}^{-1} \text{s}^{-1} \text{Hz}^{-1}$.
- ▶ $\hat{\mathbf{k}}$ is the normal to dA ; the factor $\hat{\mathbf{n}} \cdot \hat{\mathbf{k}} = \cos \Theta$ is the *projected* area seen by the beam.
- ▶ I is a *scalar* associated with a direction: it is the intensity at location $\mathbf{x}$ propagating in direction $\hat{\mathbf{n}}$.
- ▶ I is macroscopic: it sums all microscopic absorption and emission processes acting on the beam.

Why we must carry I through the calculation

- ▶ As a beam travels through the atmosphere, photons are absorbed from and emitted into it. Different parts of the planet therefore have different intensities, and I varies with frequency.
- ▶ We can *never* measure I from a specific part of an exoplanet: the planet is spatially unresolved. What we measure is a *flux*, an integral of I over the visible hemisphere.
- ▶ Nevertheless, radiative transfer is formulated in terms of I . We compute I throughout the atmosphere and integrate it only at the end to obtain observable quantities.

Two properties worth remembering (exercises 2.2 and 2.3):

- ▶ In empty space (no extinction, no emission), I is *constant along a ray*: the energy through $dA_1 d\Omega_1$ equals that through $dA_2 d\Omega_2$ because $dA_1 d\Omega_1 = dA_2 d\Omega_2$ (the étendue is conserved). Intensity does not fall off with distance.
- ▶ The *flux* of a source does fall off as $1/d^2$: the solid angle subtended by the source shrinks as $1/d^2$ while I stays fixed. There is no contradiction.

Solid angle and the mean intensity

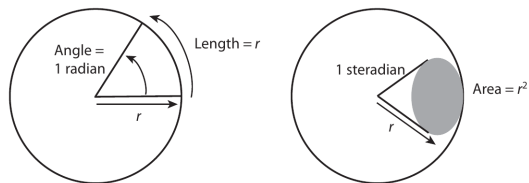


Figure 6: A radian (left) and a steradian (right). In 2D, 1 radian is the angle subtended at the center of a circle by an arc equal to the radius; in 3D, 1 steradian is the solid angle subtended at the center of a sphere of radius r by an area r^2 . Reproduced from Seager (2010), Fig. 2.2.

The full sphere subtends 4π steradians. A cone of half-opening angle θ subtends (exercise 2.4)

$$\Omega = \frac{1}{R^2} \int_0^\theta 2\pi R \sin \theta' R d\theta' = 2\pi(1 - \cos \theta),$$

which tends to 4π as $\theta \rightarrow \pi$ and to $\pi\theta^2$ (the area of a small disk over R^2) as $\theta \rightarrow 0$.

The **mean intensity** is the intensity averaged over all solid angles, the *zeroth moment* of I :

$$J(\mathbf{x}, \mathbf{v}, t) = \frac{1}{4\pi} \oint_{\Omega} I(\mathbf{x}, \hat{\mathbf{n}}, \mathbf{v}, t) d\Omega$$

For isotropic radiation $J = I$. J is what an absorbing particle “feels”: the heating rate of a gas parcel depends on J , not on the flux.

Flux: the first moment of the intensity

The quantity we actually measure from exoplanets is related to the **flux**: the *net* flow of energy through an arbitrarily oriented area dA with normal $\hat{\mathbf{k}}$, per frequency interval, per unit time. As a vector,

$$\mathbf{F}(\mathbf{x}, \nu, t) = \oint_{\Omega} I(\mathbf{x}, \hat{\mathbf{n}}, \nu, t) \hat{\mathbf{n}} d\Omega, \quad [\mathbf{F}] = \text{J m}^{-2} \text{s}^{-1} \text{Hz}^{-1},$$

defined at each point $\mathbf{x}$ of the atmosphere. In Cartesian components, $\mathbf{F} = F_i \hat{\mathbf{i}} + F_j \hat{\mathbf{j}} + F_k \hat{\mathbf{k}}$.

- In planetary-atmosphere studies, we often want the flux in *one* direction, such as toward the observer. We take one component of the vector and write only its magnitude, a scalar:

$$F_k(\mathbf{x}, \nu, t) = \frac{dE(\nu, t)}{dA d\nu dt} = \oint_{\Omega} I(\mathbf{x}, \hat{\mathbf{n}}, \nu, t) \hat{\mathbf{n}} \cdot \hat{\mathbf{k}} d\Omega$$

- The directional subscript is then usually dropped: $F(\mathbf{x}, \nu, t)$ means the flux in the direction of the observer.
- Since $\hat{\mathbf{n}} \cdot \hat{\mathbf{k}}$ changes sign across the plane of dA , F is a *net* flux: energy flowing forward minus energy flowing backward. For *isotropic* radiation the net flux through any surface vanishes, $F = 0$, although the outward flux through one hemisphere is πI (exercise 2.2).

Higher moments: K and the radiation pressure

The *second moment* of the intensity is the tensor

$$\mathbf{K}(\mathbf{x}, \nu, t) = \frac{1}{4\pi} \oint_{\Omega} I(\mathbf{x}, \hat{\mathbf{n}}, \nu, t) \hat{\mathbf{n}} \hat{\mathbf{n}} d\Omega,$$

related to the **radiation pressure tensor** by $\mathbf{P} = \frac{4\pi}{c} \mathbf{K}$, i.e. $P_{ij} = \frac{1}{c} \oint I n_i n_j d\Omega$. When K is used (Chapter 6, the Eddington approximation), only the magnitude of the component of interest is kept, a scalar, exactly as for the flux.

The hierarchy of moments (in a plane-parallel atmosphere with $\mu = \cos \theta$, Chapter 5):

$$J = \frac{1}{2} \int_{-1}^1 I d\mu, \quad H = \frac{1}{2} \int_{-1}^1 \mu I d\mu = \frac{F}{4\pi}, \quad K = \frac{1}{2} \int_{-1}^1 \mu^2 I d\mu.$$

- ▶ J : energy density ($u = 4\pi J/c$) and heating; H (or F): energy transport; K : radiation pressure ($P = 4\pi K/c$).
- ▶ For isotropic radiation $K = J/3$ and $P = u/3$, the familiar radiation-pressure relation.
- ▶ The stellar-atmosphere literature often uses the “astrophysical flux” $H = F/4\pi$ (or $\mathcal{F} = F/\pi$) because its form parallels J and K . Watch the factors of π when comparing books.

Surface flux F_S : flux emerging from the planet

For exoplanet atmospheres we are most interested in the flux emerging from the planet's “surface”: the layers from which most of the radiation originates without further interaction with gas, liquid, or solid particles. We call it the **surface flux** F_S and distinguish it from F , which can be defined in any small volume inside the atmosphere.

- ▶ F_S is the *outgoing* flux at a particular point on the surface: the magnitude of the flux vector in the outgoing direction of interest, a scalar.
- ▶ To write it down we need a coordinate system for the surface point, *and* a coordinate system for the directions $\hat{\mathbf{n}}$ of the rays leaving that point. Two choices are convenient: latitude–longitude (next slide) and colatitude (the following slide).

Coordinates on the planet. In the spherical polar system (r, θ, ϕ) , θ is the latitude and ϕ the longitude. For the planet–star problem the “equator” $\theta = 0$ is the orbital (ecliptic) plane, so $(\theta, \phi) = (0, 0)$ is the **substellar point**, the point that receives the most stellar radiation. (On Earth the equator is instead defined by the rotation axis.) Because we want only the surface intensity, we drop r and set $r = R_p$.

Latitude–longitude coordinates and $\hat{\mathbf{n}} \cdot \hat{\mathbf{k}}$

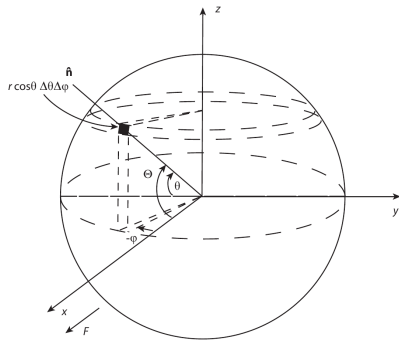


Figure 7: Spherical polar coordinates with latitude θ and longitude ϕ . For flux along the x -axis ($\hat{\mathbf{k}} = \hat{\mathbf{x}}$), $\hat{\mathbf{n}} \cdot \hat{\mathbf{k}} = \cos \Theta$. Reproduced from Seager (2010), Fig. 2.3.

The intensity now has two sets of angles: (θ, ϕ) for the surface point and (θ_n, ϕ_n) for the ray direction $\hat{\mathbf{n}}$, so $I_S(\theta, \phi, \theta_n, \phi_n, \nu, t)$.

- With the flux direction $\hat{\mathbf{k}}$ along the x -axis (toward the observer), the spherical cosine law gives

$$\hat{\mathbf{n}} \cdot \hat{\mathbf{k}} = \cos \Theta_n = \cos \theta_n \cos \phi_n$$

(exercise 2.5): a ray tilted in latitude *and* in longitude has its projection reduced by both cosines.

- The solid-angle element is $d\Omega_n = \cos \theta_n d\theta_n d\phi_n$ (the $\cos \theta_n$ is the Jacobian of latitude, since latitude is measured from the equator, not from the pole).
- Using θ_n, ϕ_n we no longer need $\hat{\mathbf{n}}$ or $\hat{\mathbf{k}}$ explicitly in the description of the intensity.

Colatitude coordinates: the symmetric choice

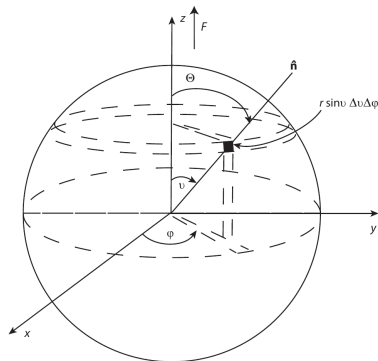


Figure 8: Spherical polar coordinates with colatitude ϑ and longitude ϕ . For flux along the z -axis, $\hat{\mathbf{n}} \cdot \hat{\mathbf{k}} = \cos \vartheta$.
Reproduced from Seager (2010), Fig. 2.4.

It is often easier to align the polar axis with the z -axis and use the **colatitude** $\vartheta = 90^\circ - \theta$ instead of the latitude. The substellar point is now at the pole and the flux is specified along z . This choice gives

$$\hat{\mathbf{n}} \cdot \hat{\mathbf{k}} = \cos \Theta_n = \cos \vartheta_n, \quad d\Omega_n = \sin \vartheta_n d\vartheta_n d\phi_n.$$

- Only one angle enters the projection factor.
- For problems symmetric in ϕ (a star at the zenith of the substellar point, an isotropically emitting surface) the azimuthal integral is trivial and gives 2π .
- This is the system used to derive the flux observed at Earth and the incident stellar flux below, with the z -axis pointing at the observer or at the star.

The surface flux in the two coordinate systems

Inserting the projection factors and solid-angle elements into the flux definition, the surface flux at the point (θ, ϕ) or (ϑ, ϕ) is

$$F_S(\theta, \phi, \mathbf{v}, t) = \int_{\phi_n=-\pi/2}^{\phi_n=\pi/2} \int_{\theta_n=-\pi/2}^{\theta_n=\pi/2} I_S(\theta, \phi, \theta_n, \phi_n, \mathbf{v}, t) \cos \phi_n \cos^2 \theta_n d\theta_n d\phi_n,$$

$$F_S(\vartheta, \phi, \mathbf{v}, t) = \int_{\phi_n=0}^{\phi_n=2\pi} \int_{\vartheta_n=0}^{\vartheta_n=\pi/2} I_S(\vartheta, \phi, \vartheta_n, \phi_n, \mathbf{v}, t) \cos \vartheta_n \sin \vartheta_n d\vartheta_n d\phi_n.$$

- The limits ($\pm\pi/2$ in latitude–longitude; 0 to $\pi/2$ in colatitude) select the *outgoing* hemisphere. In which direction does the flux travel? Out from the planet along $\hat{\mathbf{k}}$, at the location (θ, ϕ) .
- **Common shorthand.** The surface flux is usually written without reference to the surface point,

$$F_S(\mathbf{v}, t) = \int_0^{2\pi} \int_0^{\pi/2} I_S(\vartheta, \phi, \mathbf{v}, t) \cos \vartheta \sin \vartheta d\vartheta d\phi,$$

where ϑ, ϕ now refer to the *directions* of the rays and the surface element is left unspecified. Although less precise, this notation is adequate for a planet with uniform surface intensity, and we will adopt it hereafter.

The basic result: $F_S = \pi I_S$ for uniform intensity

For a planet with uniform (isotropic) surface intensity $I_S(\theta, \phi, \mathbf{v}, t) = I_S(\mathbf{v}, t)$, the intensity comes out of the integrand and the angular integral over the outgoing hemisphere gives

$$\boxed{F_S(\mathbf{v}, t) = \pi I_S(\mathbf{v}, t)} \quad \text{since} \quad \int_0^{2\pi} \int_0^{\pi/2} \cos \vartheta \sin \vartheta \, d\vartheta \, d\phi = 2\pi \cdot \frac{1}{2} = \pi.$$

- ▶ The factor is π , not 2π : rays leaving at grazing angles carry little energy *through* the surface because of the projection factor $\cos \vartheta$.
- ▶ This is why the blackbody *flux* is πB while the blackbody *intensity* is B .
- ▶ The same integral in latitude–longitude form, $\iint \cos \phi \cos^2 \theta \, d\theta \, d\phi = \pi$, gives the same answer, as it must.

Frequency-integrated (total) flux. When we integrate over all frequencies we drop the $\mathbf{v}$ argument:

$$F_S(t) = \int_0^\infty F_S(\mathbf{v}, t) \, d\mathbf{v} \quad [\text{J m}^{-2} \text{s}^{-1}].$$

Observed flux: what we measure at Earth

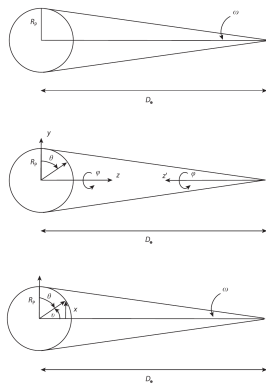


Figure 9: Derivation of the planet flux observed at Earth. Seager (2010), Fig. 2.5.

What radiative quantity can we measure at Earth? Because the planet is spatially unresolved, it appears as a point source: all radiation from its visible hemisphere is combined into a single measured flux, $F_{\oplus}(\nu, t)$.

- ▶ Recall that the surface intensity depends on the location on the planet, described by θ and ϕ . We must integrate the surface intensity from each location into a quantity we can actually measure.
- ▶ At Earth, a detector of a given area measures the energy collected from a planet that subtends a solid angle Ω , over specified frequency and time intervals.
- ▶ **The key point:** to derive $F_{\oplus}$ for a distant planet we must note that the integration over solid angle is performed *at the detector*, not at the planet surface.
- ▶ We return to the definition of flux and use the colatitude coordinate system, which is convenient here: the flux leaves the planet along the z -axis, and the z -axis points toward the detector at Earth. The angle ϕ is defined about the z -axis.

Observed flux: the solid angle subtended by the planet

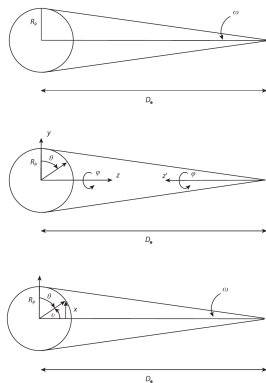


Figure 10: Top: the angle $R_p/D_{\oplus}$. Middle: the shared azimuth ϕ . Bottom: ω versus ϑ . Seager (2010), Fig. 2.5.

- From the detector at Earth, the planet subtends the solid angle

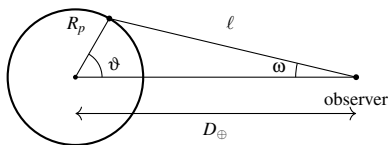
$$\Omega = \int_0^{2\pi} \int_0^{\omega_{\max}} \sin \omega \, d\omega \, d\phi \simeq \int_0^{2\pi} \int_0^{\omega_{\max}} \omega \, d\omega \, d\phi = \pi \left(\frac{R_p}{D_{\oplus}} \right)^2,$$

$$\sin \omega \simeq \omega, \quad \sin \omega_{\max} \simeq \omega_{\max} = R_p/D_{\oplus},$$

where ω is the angle measured from the planet's center as seen from Earth (top panel), running from 0 at the disk center to $R_p/D_{\oplus}$ at the limb.

- A critical point: the azimuthal coordinate ϕ on the planet's apparent disk is *identical* to the coordinate ϕ in the planet's colatitude system (middle panel). The azimuth about the line of sight is the same whether measured at the planet or at the detector.
- Each point of the planet's disk, at angular position (ω, ϕ) as seen from Earth, corresponds to one surface element at colatitude ϑ on the planet (bottom panel); the ray from that element to Earth leaves the surface at angle ϑ from the local normal.

Observed flux: relating ω to the colatitude ϑ



A surface point at colatitude ϑ lies a distance ℓ from the observer and is seen at the angle ω from the line of sight. The cosine rule for the triangle gives

$$\ell = \sqrt{R_p^2 + D_{\oplus}^2 - 2R_p D_{\oplus} \cos \vartheta},$$

and the sine rule, applied to the same triangle,

$$\ell \sin \omega = R_p \sin \vartheta \quad \Rightarrow \quad \sin \omega = \frac{R_p}{\ell} \sin \vartheta.$$

Eliminating ℓ and dividing through by $D_{\oplus}$,

$$\sin \omega = \frac{(R_p/D_{\oplus}) \sin \vartheta}{\sqrt{1 + (R_p/D_{\oplus})^2 - 2(R_p/D_{\oplus}) \cos \vartheta}}$$

which is exact. For a distant planet, $R_p/D_{\oplus} \ll 1$, the square root tends to 1 and

$$\sin \omega \simeq \frac{R_p}{D_{\oplus}} \sin \vartheta, \quad \text{so} \quad \omega \simeq \frac{R_p}{D_{\oplus}} \sin \vartheta$$

since ω is then itself small.

Observed flux: the integral at the detector

In the coordinate system of the detector, the flux is

$$F_{\oplus}(\mathbf{v}, t) = \int_0^{2\pi} \int_0^{R_p/D_{\oplus}} I_S(\vartheta, \phi, \mathbf{v}, t) \cos \omega \sin \omega \, d\omega \, d\phi,$$

where the intensity is still described in the planet's coordinates rather than in those of the detector. To convert $\cos \omega \sin \omega \, d\omega$ into the planet's coordinates, differentiate the small-angle relation of the previous slide:

$$\cos \omega \, d\omega \simeq \frac{R_p}{D_{\oplus}} \cos \vartheta \, d\vartheta,$$

and multiply by $\sin \omega \simeq (R_p/D_{\oplus}) \sin \vartheta$:

$$\cos \omega \sin \omega \, d\omega \simeq \left(\frac{R_p}{D_{\oplus}} \right)^2 \cos \vartheta \sin \vartheta \, d\vartheta$$

- The upper limit $\omega = R_p/D_{\oplus}$ corresponds to $\vartheta = \pi/2$, the limb of the visible hemisphere.
- In the same limit $D_{\oplus}$ may be taken as the distance from the surface of the star to the observer at Earth.

Substituting,

$$F_{\oplus}(\mathbf{v}, t) = \left(\frac{R_p}{D_{\oplus}} \right)^2 \int_0^{2\pi} \int_0^{\pi/2} I_S(\vartheta, \phi, \mathbf{v}, t) \cos \vartheta \sin \vartheta \, d\vartheta \, d\phi,$$

in which the angular integral is exactly the surface-flux integral.

Observed flux: the result

$$F_{\oplus}(\mathbf{v}, t) = \left(\frac{R_p}{D_{\oplus}} \right)^2 F_S(\mathbf{v}, t)$$

- ▶ $F_{\oplus}$ has the same dimensions as F_S and is smaller by the dilution factor $(R_p/D_{\oplus})^2$: the inverse-square law, now derived rather than assumed. For a Jupiter at 10 pc, $(R_p/D_{\oplus})^2 = 5 \times 10^{-17}$.
- ▶ The result is exact only for a planet with uniform surface intensity and $R_p \ll D_{\oplus}$. For a nonuniform planet, $F_{\oplus}$ is a hemisphere-weighted average with the $\cos \vartheta$ weighting: regions near the limb contribute less than regions near the center of the disk. This weighting is what phase curves and eclipse maps exploit to recover the brightness distribution.
- ▶ The surface flux and the observed flux are thus the “same” quantity up to a geometric factor, provided the planet is far away and its intensity is uniform. This is the justification for talking about the planet’s “flux” without specifying where it is evaluated, which we will often do.
- ▶ The same derivation with the roles reversed gives the stellar flux incident on the planet (next section).

Luminosity and outgoing energy

Luminosity $L(t)$ is the rate at which a planet radiates energy in *all* directions: the surface flux summed over a closed surface around the planet. For uniform flux $F_S(t)$ radiated equally everywhere, with $dA = R_p^2 \sin \theta \, d\theta \, d\phi$,

$$L(t) = \oint_A F_S(t) \, dA = R_p^2 \int_0^{2\pi} \int_0^\pi F_S(t) \sin \theta \, d\theta \, d\phi \implies \boxed{L(t) = 4\pi R_p^2 F_S(t)} \quad [\text{J s}^{-1}].$$

- For giant planets, the interior luminosity is also useful for determining the flux incident on the atmosphere's *lower* boundary. This flux is uniform across the planet and originates from gravitational energy released as the planet contracts.
- **Caution:** when observing, only *one hemisphere* is visible. For a planet that does not radiate uniformly (a hot dayside and a cold nightside) we must not use L ; instead use the energy per unit time leaving one hemisphere,

$$E_S(t) = R_p^2 \int_0^\pi \int_0^{\pi/2} F_S(\theta, \phi, t) \sin \theta \, d\theta \, d\phi \xrightarrow{\text{uniform}} E_S(t) = 2\pi R_p^2 F_S(t).$$

Energy per unit time is technically *power*; the planetary literature says “energy per unit time” or “luminosity”.

Incident flux: the star as seen from the planet

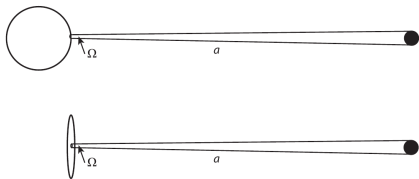


Figure 11: Incident flux on a sphere and on a disk. The star subtends a solid angle Ω as seen from each location on the planet; a is the semimajor axis. Reproduced from Seager (2010), Fig. 2.6.

Stellar radiation heats the planet, governs its global energy balance, and drives atmospheric circulation. We call it incident radiation, incident flux, or *irradiation*.

We can reuse the result for the planetary flux observed at Earth by reversing the roles: replace $R_p \rightarrow R_*$, $D_{\oplus} \rightarrow a$, and $I_S \rightarrow I_{S,*}$. The flux from the star at a point on the planet is

$$F_{\text{inc}}(\mathbf{v}, t) = \left(\frac{R_*}{a} \right)^2 \int_0^{2\pi} \int_0^{\pi/2} I_{S,*} \cos \vartheta \sin \vartheta \, d\vartheta \, d\phi.$$

For a star of uniform surface intensity (no limb darkening),

$$F_{\text{inc}}(\mathbf{v}, t) = \left(\frac{R_*}{a} \right)^2 \pi I_{S,*}(\mathbf{v}, t) = \left(\frac{R_*}{a} \right)^2 F_{S,*}(\mathbf{v}, t)$$

This is the flux at the **substellar point**, where the star is at the zenith.

Incident energy: integrating over the dayside

The total incident energy per unit time is *not* $2\pi R_p^2 F_{\text{inc}}$: only the substellar point receives the full flux. A point at angle Θ from the substellar point receives $F_{\text{inc}} \cos \Theta$ because the surface element is tilted (this is why Earth's poles are colder than the equator). Integrating over the illuminated hemisphere,

$$E_{\text{inc}}(\mathbf{v}, t) = \left(\frac{R_*}{a}\right)^2 F_{\text{S},*}(\mathbf{v}, t) \int_0^{2\pi} \int_0^{\pi/2} R_p^2 \cos \vartheta \sin \vartheta \, d\vartheta \, d\phi \quad \Rightarrow \quad E_{\text{inc}}(\mathbf{v}, t) = \pi R_p^2 \left(\frac{R_*}{a}\right)^2 F_{\text{S},*}(\mathbf{v}, t)$$

and, integrated over all wavelengths, $E_{\text{inc}}(t) = \pi R_p^2 (R_*/a)^2 F_{\text{S},*}(t)$.

- ▶ The $\cos \vartheta$ integral turns the hemisphere ($2\pi R_p^2$) into the **cross-sectional disk** πR_p^2 : the planet intercepts starlight with its projected area. This is the “disk” in Fig. 2.6.
- ▶ Sanity check with $L_* = 4\pi R_*^2 F_{\text{S},*}$: the stellar flux at distance a is $L_*/4\pi a^2 = (R_*/a)^2 F_{\text{S},*}$, and the planet intercepts πR_p^2 of it. For Earth, $L_{\odot}/4\pi a^2 = 1361 \text{ W m}^{-2}$ (the solar constant).

The star as a point source: plane-parallel incident radiation

For radiative-transfer calculations at a given surface element, the star is far enough away that only rays from *one* direction are incident. We then describe the incident stellar intensity with delta functions,

$$I_*(\vartheta, \phi, \nu, t) = I_0 \delta(\vartheta - \vartheta_0) \delta(\phi - \phi_0),$$

where (ϑ_0, ϕ_0) is the direction of the star from the local surface normal; this direction differs from one surface element to another. The incident flux at the element is then

$$F_{\text{inc}}(\vartheta, \phi, \nu, t) = \left(\frac{R_*}{a}\right)^2 \int_0^{2\pi} \int_0^{\pi/2} I_0 \delta(\vartheta - \vartheta_0) \delta(\phi - \phi_0) \cos \vartheta \sin \vartheta \, d\vartheta \, d\phi = \left(\frac{R_*}{a}\right)^2 \cos \vartheta_0 \sin \vartheta_0 I_0.$$

- ▶ The $\cos \vartheta_0$ is the projection factor; $\sin \vartheta_0$ comes from the solid-angle Jacobian and the normalization of the delta functions in (ϑ, ϕ) .
- ▶ This “plane-parallel beam” form is the upper boundary condition of the plane-parallel radiative transfer equation (Chapter 5): $I(0, \mu, \nu) = I_* \delta(\mu - \mu_0)$ for incoming rays.
- ▶ *When is the point-source approximation valid?* (exercise 2.7) When the star’s angular radius R_*/a is small compared to the angular scales over which the atmosphere’s response varies. For a hot Jupiter at $a = 0.05$ AU, $R_*/a \simeq 0.09$ rad $\simeq 5^\circ$, so near the terminator the finite size of the star (and its limb darkening) already matters.

Blackbody intensity and blackbody flux

A **blackbody** is a perfect radiator: it absorbs all incident radiation and re-emits it with a spectrum that depends only on its temperature T . Blackbody radiation is *isotropic* (no $\hat{\mathbf{n}}$ -dependence). Its intensity is the **Planck function**

$$B(T, \nu) = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/k_B T} - 1} \quad [\text{J m}^{-2} \text{ sr}^{-1} \text{ s}^{-1} \text{ Hz}^{-1}],$$

the same dimensions as I . Since T varies with location (and possibly time) we may also write $B(\mathbf{x}, \nu)$.

- **Per wavelength.** Blackbody radiation per wavelength bin $d\lambda$ requires the conversion $d\nu = -(c/\lambda^2)d\lambda$ (the sign is absorbed into the direction of $d\lambda$):

$$B(T, \lambda) = \frac{2hc^2}{\lambda^5} \frac{1}{e^{hc/\lambda k_B T} - 1}, \quad B_\lambda d\lambda = B_\nu d\nu.$$

B_λ and B_ν peak at *different* locations (next slide), a common source of confusion (Heng, problem 2.8.4).

- **Blackbody flux.** Because B is isotropic, the flux outward through one hemisphere is

$$F_S(T, \nu) = \pi B(T, \nu)$$

We always use B for blackbody intensity and πB for blackbody flux.

Blackbody fluxes across the relevant temperatures

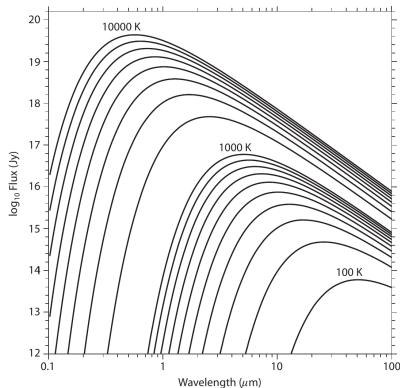


Figure 12: Blackbody surface fluxes πB (in $\text{Jy} = 10^{-26} \text{ W m}^{-2} \text{ Hz}^{-1}$), spaced by 100 K from 100 to 1000 K and by 1000 K from 1000 to 10,000 K. Seager (2010), Fig. 2.7.

- ▶ Planet-hosting stars typically have $T_{\text{eff}} \simeq 3000\text{--}6000 \text{ K}$ (the Sun may be represented by a 5750 K blackbody). Known planets have $T_{\text{eff}} \simeq 60$ to more than 2000 K (Earth is approximately a 300 K blackbody).
- ▶ Both the *magnitude* and the *frequency of the peak* of the blackbody flux increase with temperature; the curves never cross. A hotter body is brighter at *every* wavelength.
- ▶ Stellar and planetary atmospheres can be *approximated* as blackbodies, but a blackbody is a highly idealized construct; the actual spectra depart from it in bands and lines, which is where the information about composition lies.

The two sides of the Planck function

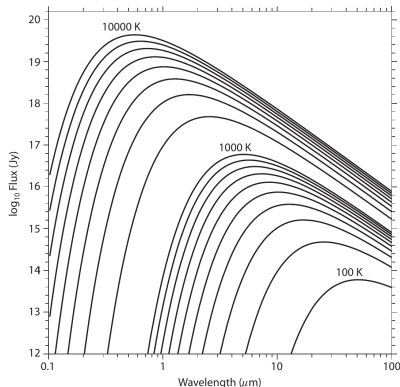


Figure 13: As in the previous slide; note the common slope on the long-wavelength side. Seager (2010), Fig. 2.7.

- **Rayleigh–Jeans side** ($h\nu \ll k_{\text{B}}T$, $\lambda \gg \lambda_{\text{max}}$): $B \simeq 2\nu^2 k_{\text{B}}T/c^2$. On this logarithmic plot it is a straight line of slope -2 in λ (slope $+2$ in ν), with the temperature entering only *linearly*. This is why the mid-infrared planet-to-star contrast is $\propto T_p/T_*$ rather than exponentially small (Lecture 1, and next lecture).
- **Wien side** ($h\nu \gg k_{\text{B}}T$): $B \simeq (2h\nu^3/c^2)e^{-h\nu/k_{\text{B}}T}$, exponentially sensitive to T . A planet's flux at wavelengths shortward of its peak is negligible, which is why planets are invisible in the optical except by reflected light.
- The **peak** moves as $\lambda_{\text{max}} \propto 1/T$ (Wien's law, next slide); the **total** flux grows as T^4 (Stefan–Boltzmann, Section 3.3).

Wien's law: where does a blackbody peak?

Setting $\partial B(T, \lambda)/\partial \lambda = 0$ or $\partial B(T, \nu)/\partial \nu = 0$ (exercise 2.6) gives **Wien's displacement law** in the two forms

$$\lambda_{\max} T = 2898 \mu\text{m K}$$

$$\frac{\nu_{\max}}{T} = 5.88 \times 10^{10} \text{ Hz K}^{-1}$$

The two peaks do *not* represent the same spectral location: $\lambda_{\max} \nu_{\max} \neq c$, because B_{λ} and B_{ν} are densities per unit wavelength and per unit frequency, and $d\nu/d\lambda = -c/\lambda^2$ redistributes the weight. (Wien derived the law from thermodynamics; the numerical constants come from the transcendental equations $x = 5(1 - e^{-x})$ and $x = 3(1 - e^{-x})$, with $x = hc/\lambda k_B T$ or $h\nu/k_B T$.)

object	T (K)	$\lambda_{\max}$ of B_{λ}	$c/\nu_{\max}$ of B_{ν}
Sun (G2)	5750	$0.50 \mu\text{m}$	$0.88 \mu\text{m}$
M star	3500	$0.83 \mu\text{m}$	$1.46 \mu\text{m}$
hot Jupiter	1500	$1.9 \mu\text{m}$	$3.4 \mu\text{m}$
Earth	300	$9.7 \mu\text{m}$	$17 \mu\text{m}$
Jupiter	125	$23 \mu\text{m}$	$41 \mu\text{m}$

Consequences of Wien's law

- ▶ The Sun's B_λ peaks in the green; its B_ν peaks in the near-infrared. Neither is “the” color of the Sun; the choice of variable is a convention. When a paper quotes “the peak of the spectrum,” check which variable is meant.
- ▶ Planets radiate in the thermal infrared (5–50 μm), stars in the visible and near-infrared: the two components of a planet's spectrum, reflected starlight and thermal emission, are well separated in wavelength, except for the hottest planets ($T \gtrsim 2000\text{ K}$), whose thermal emission reaches into the near-infrared and even the red optical.
- ▶ Detecting a planet's thermal emission means observing near its Wien peak or beyond: 3–10 μm for hot Jupiters (Spitzer, JWST), 10–25 μm for temperate terrestrial planets (the 10 μm window for Earth analogs).
- ▶ Conversely, the stellar spectrum at the planet's thermal wavelengths is on the Rayleigh–Jeans tail, faint and smooth, which is what makes secondary eclipses measurable.
- ▶ Wien's law also sets the *photochemistry*: an M dwarf with $\lambda_{\text{max}} \simeq 0.8\text{--}1\text{ }\mu\text{m}$ emits much less near-UV than the Sun, so photolysis of H_2O , CH_4 , and O_3 proceeds differently on its planets, although flares and the chromospheric far-UV can compensate.

The Lambert surface: a surface of equal brightness

A **Lambert surface** scatters intensity *isotropically* (equally in all directions). It is the standard idealization of the reflectivity of planetary bodies, and it clarifies the difference between the intensity *emanating* from an object and the intensity *measured* by a distant observer.

- ▶ “Equal intensity in all directions” means the apparent surface brightness of an area element is the same from every viewing angle.
- ▶ Example: a sheet of white paper illuminated from above appears equally bright from any direction. Likewise, the spot cast by a light bulb on a white wall maintains its surface brightness when viewed from different angles.
- ▶ Contrast with a mirror (specular reflection) or with ice crystals that strongly backscatter: the brightness then depends strongly on the viewing direction.

We now show that a plane Lambert surface, illuminated along its normal, has the same apparent brightness from any direction, and see *why* this is consistent with fewer photons leaving at slant angles.

Two observers of a Lambert surface

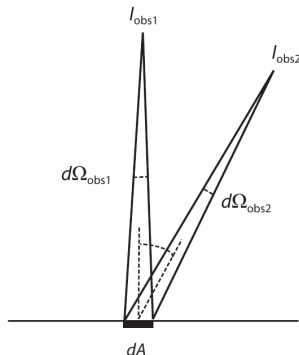


Figure 14: Intensity observed by a distant viewer. The solid angle subtended by dA is smaller at observer 2 than at observer 1. Reproduced from Seager (2010), Fig. 2.8.

Consider a plane Lambert surface illuminated along its normal. Observer 1 looks along the normal, observer 2 at an angle ϑ from it; both view the same differential area dA from a distance much larger than the size of the surface element. From the definition of intensity,

$$I_{\text{obs1}} = \frac{dE_{\text{obs1}}}{dA d\Omega_{\text{obs1}} \hat{\mathbf{n}}_{\text{obs1}} \cdot \hat{\mathbf{k}}},$$

$$I_{\text{obs2}} = \frac{dE_{\text{obs2}}}{dA d\Omega_{\text{obs2}} \hat{\mathbf{n}}_{\text{obs2}} \cdot \hat{\mathbf{k}}},$$

with, by the definition of the coordinate system, $\hat{\mathbf{n}}_{\text{obs1}} \cdot \hat{\mathbf{k}} = 1$ and $\hat{\mathbf{n}}_{\text{obs2}} \cdot \hat{\mathbf{k}} = \cos \vartheta$.

- **Geometry:** the solid angle subtended by dA at observer 2 is smaller than at observer 1 by the factor $\cos \vartheta$ (it approaches zero as $\vartheta \rightarrow 90^\circ$, when the element is seen edge-on).

Why a Lambert surface looks equally bright from all angles

- **Physics (Lambert's cosine law):** for isotropic scattering the energy leaving per unit *projected* area is the same in every direction, so the energy (or number of photons) emerging from dA toward observer 2 drops as $\cos \vartheta$ away from the normal:

$$dE_{\text{obs2}} = \cos \vartheta \, dE_{\text{obs1}}.$$

Combining these two results, observer 2 measures

$$I_{\text{obs2}} = \frac{dE_{\text{obs1}} \cos \vartheta}{dA \, d\Omega_{\text{obs2}} \cos \vartheta} \implies \boxed{I_{\text{obs1}} = I_{\text{obs2}}}$$

by comparison with the expression for observer 1.

- For a plane Lambert surface the intensity, i.e. the *brightness*, appears the same from all directions: the smaller number of photons emerging in the slant direction is exactly compensated by the smaller subtended angle of the area element.
- The *flux* received by observer 2 does decrease as $\cos \vartheta$; only the intensity is invariant. This is why a Lambert disk viewed obliquely looks just as bright but appears foreshortened.
- The same reasoning applied to a *sphere* gives the geometric albedo $2/3$ and the phase curve of a Lambert sphere (Chapter 3): each surface element is Lambertian, but the elements near the limb are lit obliquely, so they scatter fewer photons per unit area.

Chapter 2 summary and the SI radiometry vocabulary

What we defined.

- ▶ Intensity I (energy per area, solid angle, time, frequency); mean intensity J ; flux $F = \oint I \hat{\mathbf{n}} \cdot \hat{\mathbf{k}} d\Omega$; second moment K .
- ▶ Surface flux $F_S = \pi I_S$ for uniform intensity; observed flux $F_{\oplus} = (R_p/D_{\oplus})^2 F_S$.
- ▶ Luminosity $L = 4\pi R_p^2 F_S$; hemispheric energy $E_S = 2\pi R_p^2 F_S$.
- ▶ Incident flux $F_{\text{inc}} = (R_*/a)^2 F_{S,*}$; incident energy $E_{\text{inc}} = \pi R_p^2 (R_*/a)^2 F_{S,*}$.
- ▶ Blackbody: $B(T, \nu)$, flux πB .
- ▶ Lambert surface: isotropic scattering, equal brightness from all directions.

SI radiometry names (Table 2.1) differ from the astronomical usage; the correspondence is:

SI quantity	unit	our name
radiant flux	W	luminosity, L
radiant intensity	W sr ⁻¹	—
radiance	W sr ⁻¹ m ⁻²	intensity (total)
spectral radiance	W sr ⁻¹ m ⁻² Hz ⁻¹	intensity I_{ν}
irradiance	W m ⁻²	incident flux
radiant exitance	W m ⁻²	surface flux F_S
spectral irradiance	W m ⁻² Hz ⁻¹	flux F_{ν}

Atmospheric scientists use *radiance/irradiance* where astronomers use *intensity/flux*; the underlying physics is the same.

References (Chapter 2)

- ▶ Seager, S. 2010, *Exoplanet Atmospheres: Physical Processes* (Princeton University Press), Chapter 2.
- ▶ Heng, K. 2017, *Exoplanetary Atmospheres: Theoretical Concepts and Foundations* (Princeton University Press), problem 2.8.4 (the forms of the Planck function).
- ▶ Rybicki, G. B. and Lightman, A. P. 1986, *Radiative Processes in Astrophysics* (Wiley), Chapter 1.
- ▶ Mihalas, D. 1978, *Stellar Atmospheres*, 2nd ed. (Freeman).
- ▶ Shu, F. 1991, *The Physics of Astrophysics, Vol. I: Radiation* (University Science Books).
- ▶ McCluney, W. R. 1994, *Introduction to Radiometry and Photometry* (Artech House).
- ▶ Turnbull, M. C., et al. 2006, *ApJ* **644**, 551; Pearl, J. C. and Christensen, P. R. 1997, *JGR* **102**, 10875 (Earth's visible and mid-IR spectra).

Homework (due: 2026 October 6, Tuesday)

Seager, *Exoplanet Atmospheres*, Chapter 2, exercises 2–5.

- Q1.** Explain the meaning of isotropic radiation. Show that for isotropic radiation the flux integrated over a hemispheric solid angle is $F = \pi I$, but that the flux integrated over the full solid angle is $F = 0$. Show that for isotropic radiation $I = J$.
- Q2.** Show that the intensity I does not depend on distance in a medium with no extinction or emission. Show that the flux F follows the inverse square law, $F \sim 1/d^2$, where d is the distance from the source, and that this distance dependence is not in conflict with the constancy of I .
- Q3.** Show that the general expression for the solid angle (in steradians) is

$$\Omega = \frac{1}{R^2} \int_0^\theta 2\pi R \sin \theta' R d\theta' = 2\pi (1 - \cos \theta).$$

What is the large-angle limit as $\theta \rightarrow \pi$? What is the small-angle limit as $\theta \rightarrow 0$?

- Q4.** Show that $\cos \Theta = \cos \phi \cos \theta$ in the latitude–longitude spherical polar coordinate system (Seager, Fig. 2.3).